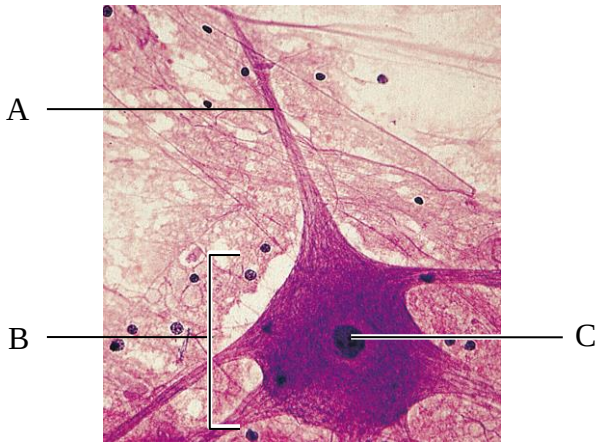
Structured questions

|!|BCELC0170000001|!|

The photomicrograph shows part of a neurone.



(a) Name structures A, B and C. (3 marks)

(b) Name the junction where two neurones meet. (1 mark)

(c) Describe how a nerve impulse is transmitted from one neurone to another. (4 marks)

##

(a) A: nerve fibre (1)
B: nucleus (1)
C: cell body (1)

(b) Synapse (1)

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(c) When a nerve impulse arrives at a synaptic knob, it stimulates the synaptic vesicles to release neurotransmitters. (1)

Neurotransmitters diffuse across the synaptic cleft. (1)

Neurotransmitters bind to specific receptor sites on the dendron of the next neurone. (1)

The dendron is stimulated to generate a nerve impulse to continue with the transmission. (1)

(8 marks)

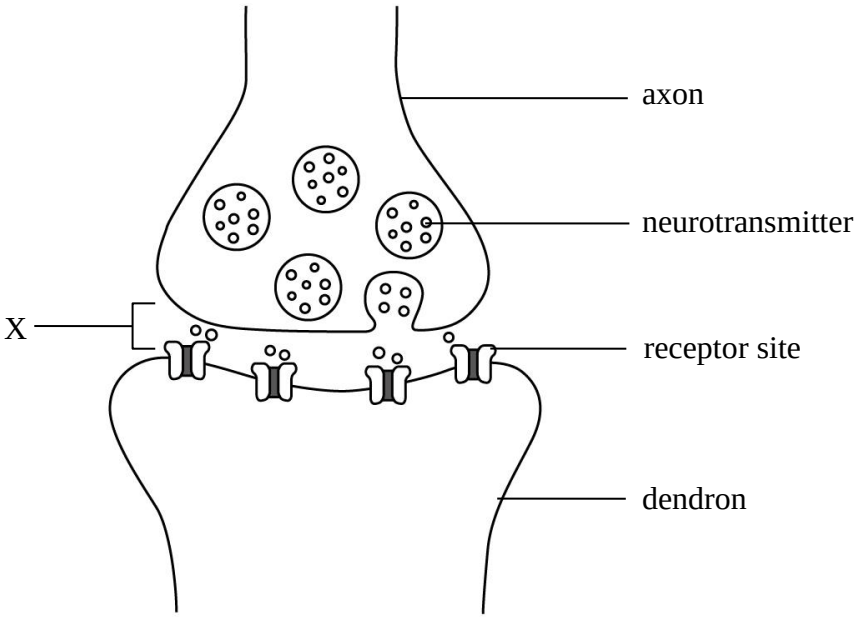
##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	4

|!|BCELA0170000002|!| ★★ **STSE** **SI**

(a) Describe ***two*** roles of synapses in the nervous system. (2 marks)

(b) The diagram below shows a synapse formed by two adjacent neurones.



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(i) Name gap X. (1 mark)

(ii) What is the difference between the transmission of signals along a nerve fibre and across a synapse? (2 marks)

(c) Parkinson's disease is a disorder of the central nervous system. It results from degeneration of neurones in a certain area of the brain and deficiency of a neurotransmitter. Symptoms of Parkinson's disease include shaking and rigid muscles.
With reference to the roles of neurones and neurotransmitters, suggest reasons to explain the above symptoms of Parkinson's disease. (2 marks)

(d) A scientist has isolated a new compound from the rat brain. List ***two*** of the criteria that the scientist should look for to confirm that the compound is a neurotransmitter. (2 marks)

##

(a) Synapses ensure that nerve impulses are transmitted in one direction only. (1)
Synapses allow a neurone to communicate with many other neurones. (1)

(b) (i) Synaptic cleft (1)

(ii) The nerve impulses transmitted along a nerve fibre are electrical signals. (1)
The neurotransmitters diffusing across the synaptic cleft are chemicals. (1)

(c) The neurones in that particular area are responsible for producing neurotransmitters (1)
and this type of neurotransmitters is involved in coordinating voluntary responses / muscle contraction / muscle coordination. (1)

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(d) Any ***two*** of the following: (2)

- It has to be synthesized in presynaptic neurones.
- It has to be found in vesicles presynaptic neurones.
- It has to be released by presynaptic stimulation.
- It can bind specifically to receptor sites in the postsynaptic neurone.
- It can stimulate the postsynaptic neurone to generate a nerve impulse.

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(c)	Applying concepts	2
(d)	Critical thinking	2

|!|BCELC0170000003|!|

Figure 1 shows a neurone, X. Figure 2 shows a section of the spinal cord.

Figure 1

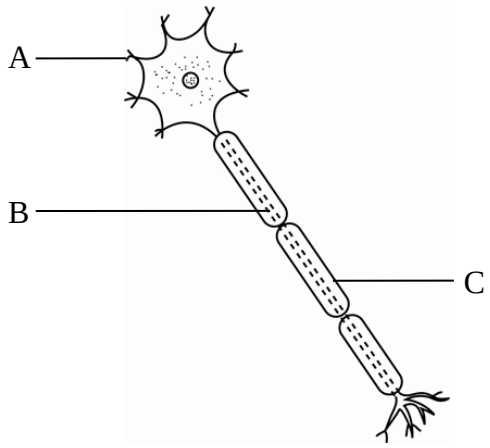
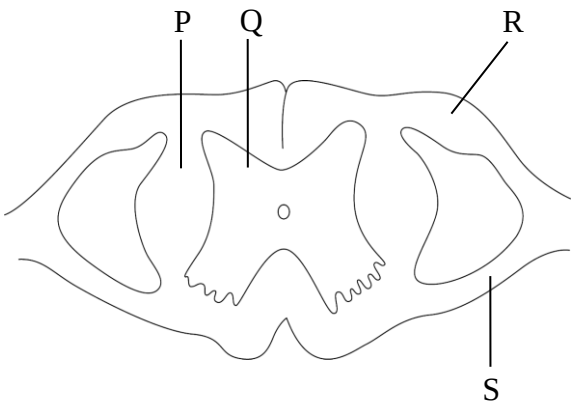


Figure 2



(a) (i) Name structures A and B. (2 marks)

(ii) State ***one*** functional difference between A and B. (1 mark)

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(b) Name structure C and state ***one*** of its functions. (2 marks)

(c) Using the letters in Figure 2, state where neurone X can be found in the spinal cord. (1 mark)

(d) What are the functions of the spinal cord? (2 marks)

##

(a) (i) A: dendron / dendrite (1)
B: axon (1)
(ii) A dendron / dendrite transmits nerve impulses towards the cell body while an axon transmits nerve impulses away from the cell body. (1)

(b) Myelin sheath (1)
It protects nerve fibres. / It insulates nerve fibres. / It speeds up the transmission of nerve fibres. (1)

(c) S (1)

(d) It transmits nerve impulses between the brain and other parts of the body. (1)
It is the reflex centre for many reflex actions involving the trunk and the limbs. (1)

(8 marks)

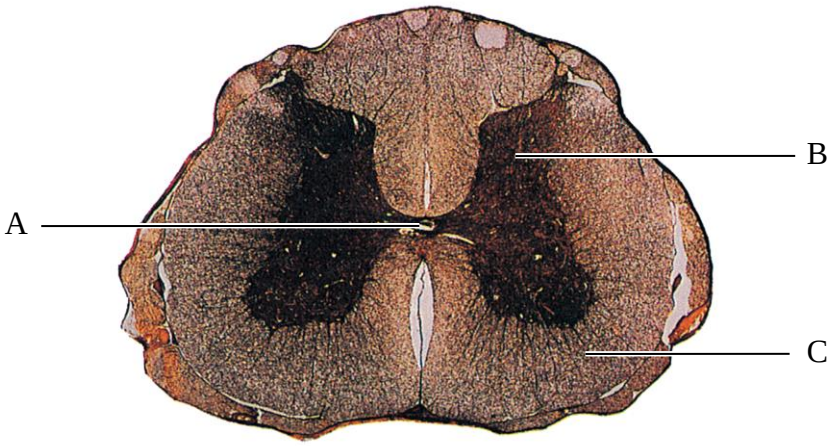
##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Understanding basic concepts	1
(b)	Analysing data	1
	Understanding basic concepts	1
(c)	Analysing data	1
(d)	Understanding basic concepts	2

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|!|BCELB0170000004|!| ★ **STSE**

The photomicrograph below shows a section of the spinal cord.



(a) (i) Name structure A. (1 mark)

(ii) Name the fluid in structure A and state ***one*** of its functions. (2 marks)

(b) (i) Name regions B and C. (2 marks)

(ii) What do regions B and C mainly consist of? (2 marks)

(c) To check the functioning of the spinal cord, a doctor may tap on the patient's hanging leg just below the knee cap. Explain why this works. (2 marks)

##

(a) (i) Central canal (1)

(ii) Cerebrospinal fluid (1)

It provides nutrients and oxygen to the neurones and removes waste from them. / It acts as a shock absorber. / It provides mechanical support to the spinal cord. (1)

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(b) (i) B: grey matter (1)
C: white matter (1)
(ii) Region B consists of the cell bodies of interneurons and motor neurons. (1)
Region C consists of myelinated nerve fibres. (1)

(c) The spinal cord serves as the reflex centre for the knee jerk reflex. / The spinal coordinates the knee jerk reflex. (1)
If the spinal cord is injured / damaged, the knee jerk reflex may not happen. (1)

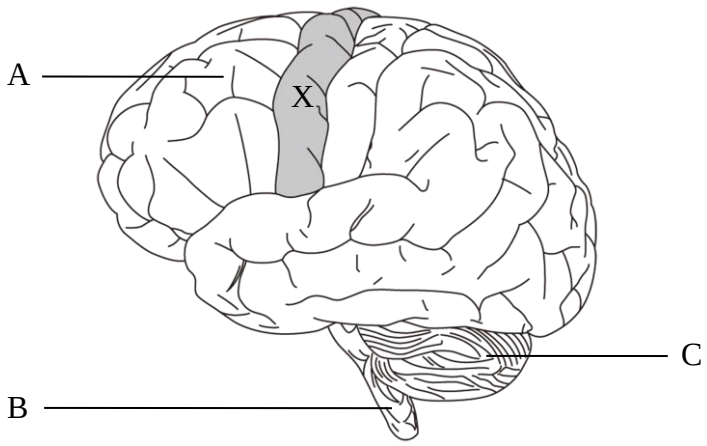
(9 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	2
(b)(i)	Analysing data	2
(b)(ii)	Understanding basic concepts	2
(c)	Applying concepts	2

|!|BCELA0170000005|!| ★★ **STSE**

The diagram below shows a sectional view of the human brain.



(a) (i) What is grey matter and white matter in the brain? (2 marks)

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(ii) Compare the distribution of grey matter and white matter in parts A, B and C of the brain. (3 marks)

(b) (i) A stroke occurs when the blood supply to part of the brain is disrupted causing brain cells to die. Explain why a lack of blood flow to part of the brain results in death of brain cells. (2 marks)

(ii) A patient suffered a stroke, which left him with weakness down the right side of his body. Region X of his brain was diagnosed to be damaged. What is region X? (1 mark)

##

(a) (i) Grey matter consists of the cell bodies of neurons (motor neurones and interneurones). (1)
White matter consists of nerve fibres of neurones. (1)
(ii) Both A and C have the outer layer made up of grey matter and the inner layer made up of white matter. (2)
B has the outer layer made up of white matter and the inner layer made up of grey matter. (1)
(b) (i) The brain cells cannot obtain sufficient oxygen and glucose for respiration. (1)
Metabolic wastes accumulate and become toxic to the cells. (1)
(ii) Motor area (1)

(8 marks)

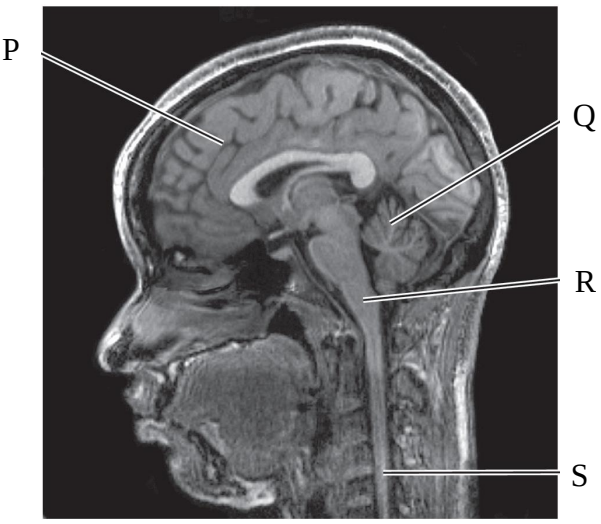
##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Analysing data	3
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	1

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|!|BCELB0170000006|!| ★

(a) The photograph below is an MRI scan of the head.



With reference to the MRI scan, complete the table below by putting in the correct letter for each description. (3 marks)

Description of part of the brain	Letter
responsible for controlling the heart rate	
responsible for memory	
reflex centre for reflex actions involving the trunk and the limbs	

(b) How is the brain protected from injury? (3 marks)

(c) A man was injured in a traffic accident and became paralyzed from the neck down. Suggest how this might have occurred. (3 marks)

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##

(a)

Description of part of the brain	Letter
responsible for controlling the heart rate	R
responsible for memory	P
reflex centre for reflex actions involving the trunk and the limbs	S

(3)

(b) The brain is enclosed by the cranium. (1)

The brain is covered by three membranes (meninges). (1)

The space between the inner and middle membranes is filled with the cerebrospinal fluid which acts as a shock absorber. (1)

(c) The spinal cord near the neck might have been damaged in the accident. (1)

The spinal cord could no longer transmit nerve impulses between the brain and the parts of the body below the neck. (1)

Muscles (effector) below the neck could no longer receive nerve impulses from the brain. (1)

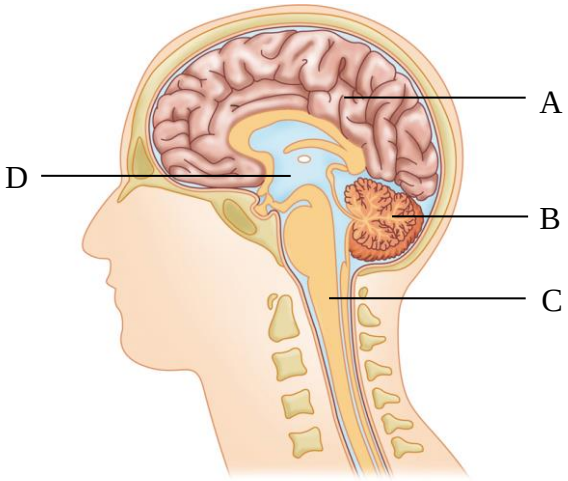
(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	3
(c)	Applying concepts	3

|!|BCELB0170000007|!| ★ **Cross topic**

The diagram below shows a section through the human head.



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(a) Identify A, B and C, and state ***one*** function for each. (6 marks)

(b) D is filled with a fluid. This fluid is formed by filtration of the blood. The fluid is similar to the blood but contains no blood cells or plasma proteins. It bathes brain tissues, removing metabolic wastes before returning into the bloodstream.

(i) Name this fluid. (1 mark)

(ii) With reference to water potential, explain how the fluid is returned into the bloodstream. (3 marks)

##

(a) A: cerebrum / cerebral hemisphere (1)
The cerebrum controls voluntary actions. / The cerebrum is responsible for memory / thinking / judgement. (1)
B: cerebellum (1)
The cerebellum coordinates muscular movements. / The cerebellum control balance of the body. (1)
C: medulla oblongata (1)
The medulla oblongata is the reflex centre for many reflex actions in the head. / The medulla oblongata controls involuntary actions. (1)

(b) (i) Cerebrospinal fluid (1)

(ii) Due to the retention of plasma proteins in the blood, (1)
the cerebrospinal fluid has a higher water potential than that of the blood. (1)
Hence, water from the cerebrospinal fluid returns into the bloodstream down the water potential gradient by osmosis. (1)

(10 marks)

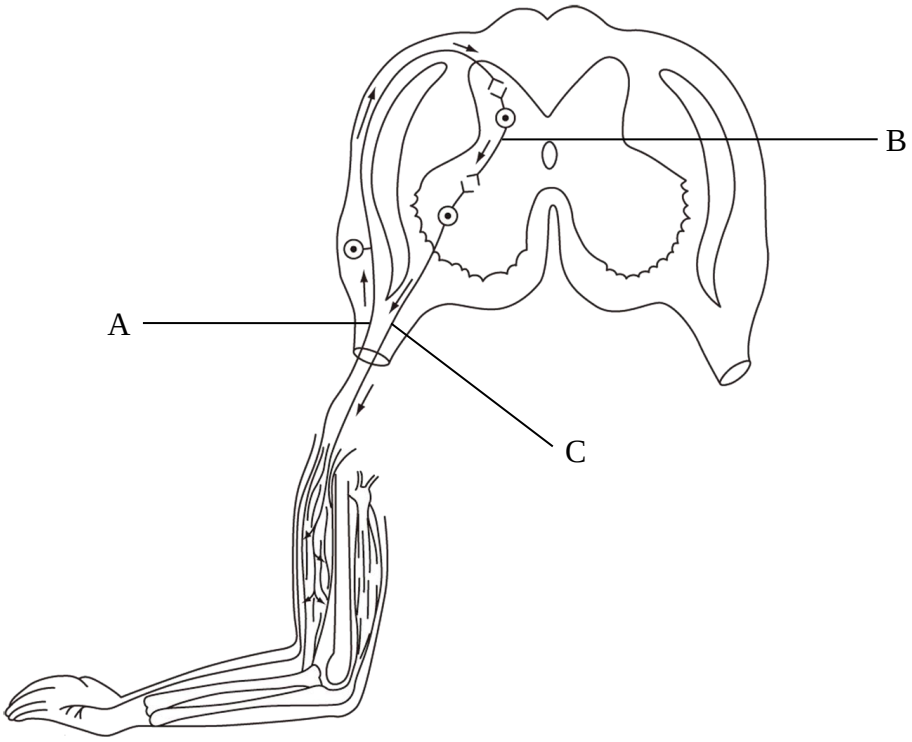
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##

	Skills assessed	Marks
(a)	Analysing data	3
	Understanding basic concepts	3
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	3

|!|BCELB0170000008|!| ★ **Cross topic**

When a hand accidentally touches a sharp object, a reflex action occurs. The biceps muscle contracts, causing the arm to be flexed and the hand is pulled away. The diagram below shows the nervous pathway involved in the reflex action.



(a) Explain the importance of this kind of reflex action. (2 marks)

(b) Identify neurones A, B and C. (3 marks)

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(c) Explain why the sensation of pain usually lags behind the response in such a reflex action. (2 marks)

(d) If the ventral root of the spinal nerve was damaged, how would the sensation of pain and the response be affected? (2 marks)

✂

(e) After the reflex action, the arm is straightened by a voluntary action. Describe how the arm is straightened after the reflex action has occurred. (4 marks)

##

(a) They are fast / automatic responses (1)
that protect the body from injury / danger. (1)

(b) A: sensory neurone (1)
B: interneurone (1)
C: motor neurone (1)

(c) The pathway through which nerve impulses are transmitted to the cerebrum is longer than the reflex arc. (1)
More synapses are involved. (1)

(d) The person would not feel any pain (1)
and would not withdraw his/her hand. (1)

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(e) The cerebrum makes the decision and generates nerve impulses. (1)
The nerve impulses are transmitted along the motor neurone to the triceps muscle. (1)
The triceps muscle is stimulated to contract (1)
and the biceps muscle relaxes to straighten the arm. (1)

(13 marks)

##

	Skills assessed	Marks
(a)	Critical thinking	2
(b)	Analysing data	3
(c)	Applying concepts	2
(d)	Applying concepts	2
(e)	Applying concepts	4

|!|BCELB0170000009|!| ★

A woman accidentally touched a hot pan. She immediately withdrew her hand from the hot pan.

(a) Describe how this reflex action was brought about. (5 marks)

(b) The reflex action did not involve the brain. How could the woman feel the pain? (2 marks)

(c) Apart from being very fast, state ***one*** other feature of ***all*** reflex actions. (1 mark)

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##

(a) The pain receptors in the skin of the hand were stimulated. (1)

The pain receptors generated nerve impulses. (1)

Nerve impulses were transmitted along a sensory neurone the spinal cord. (1)

In the spinal cord, nerve impulses were transmitted to an interneruone and then to a motor neurone. (1)

The motor neurone carried nerve impulses to the arm muscle to cause muscle contraction and withdrawal of the hand. (1)

(b) Nerve impulses are also carried to the cerebrum through another nervous pathway. (1)

The cerebrum interprets the nerve impulses to give the sensation of pain. (1)

(c) They are inborn. / They are not under voluntary control. / The cerebrum is not involved. / They always involve a receptor. / The same stimulus always results in the same response. (1)

(8 marks)

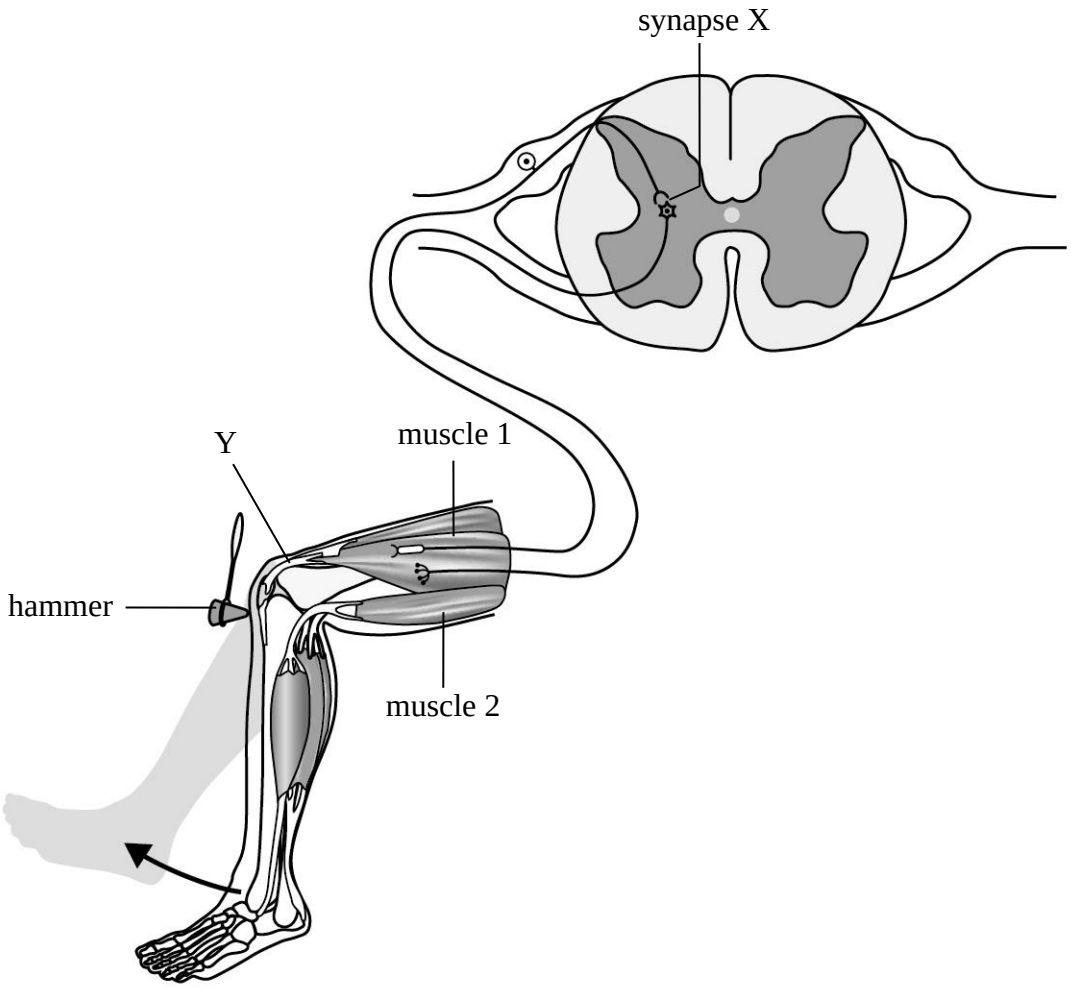
##

	Skills assessed	Marks
(a)	Understanding basic concepts	5
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1

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|!|BCELA0170000010|!| ★★ **Cross topic**

The diagram below shows the nervous pathway and the related structures involved in the knee jerk reflex.



When the tendon just below the knee cap is tapped, the lower leg automatically kicks forwards.

(a) Name the structure that generates nerve impulses to start this reflex. (1 mark)

_____

(b) Name, in their correct order, the neurones involved in this reflex. (2 marks)

_____

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(c) Describe the events that take place at synapse X during this reflex. (3 marks)

✂

(d) Muscles 1 and 2 are antagonistic muscles. Explain what this mean with reference to their action in the knee jerk reflex. (2 marks)

✂

(e) Explain how the movement of the leg would be affected if structure Y is broken. (3 marks)

##

(a) Stretch receptor (1)

(b) Sensory neurone → motor neurone (2)
(0 mark for wrong sequence, even with correct spelling)

(c) The arrival of a nerve impulse at the synaptic knob of the presynaptic neurone (sensory neurone) stimulates the release of neurotransmitters from synaptic vesicles. (1)
Neurotransmitters diffuse across the synaptic cleft. (1)
Neurotransmitters bind to receptor sites on the dendron of the postsynaptic neurone (motor neurone), stimulating the dendron to generate a nerve impulse. (1)

(d) Antagonistic muscles are muscle pair work in pairs, where one muscle contracts and the other relaxes. (1)
In the knee jerk reflex, muscle 1 contracts and muscle 2 relaxes to allow the lower leg to kick forwards. (1)

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(e) Structure Y connects muscle 1 to the bone at the knee joint. (1)

If structure Y is broken, the force generated by the contraction of muscle 1 cannot be transmitted to the bone. (1)

The leg cannot be straightened at the knee. (1)

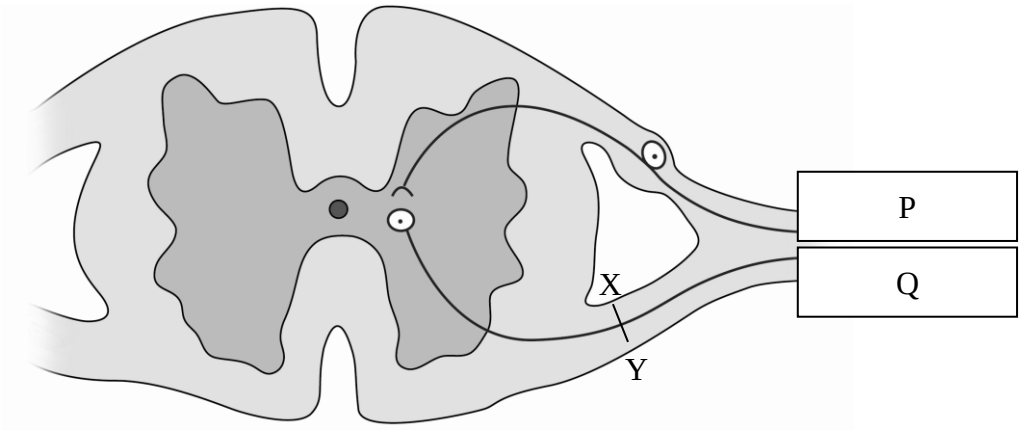
(11 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	3
(d)	Understanding basic concepts	1
	Applying concepts	1
(e)	Understanding basic concepts	1
	Applying concepts	2

|!|BCELB0170000011|!| ★

A boy was asked to sit with his legs crossed. His hanging leg kicked when the tendon below the kneecap is tapped. The diagram below shows the nervous pathway of this reflex action in the spinal cord.



(a) What are P and Q? (2 marks)

(b) Label the two neurones in the above diagram. (2 marks)

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(c) Explain what would happen to the reflex if the nerve was cut across at X-Y.

(3 marks)

(d) Give ***one*** other reflex action which occurs in the human body and state its purpose.

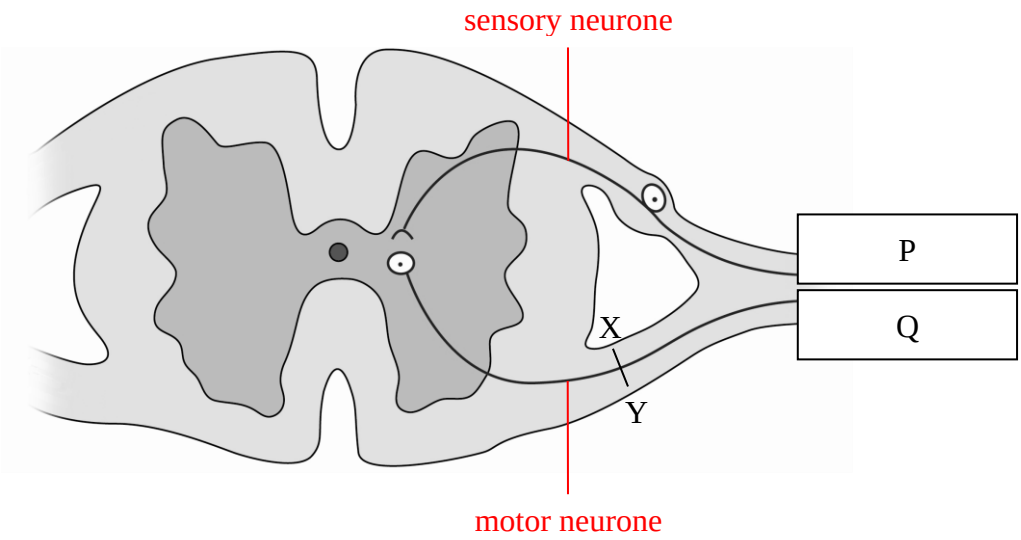
(2 marks)

##

(a) P: (stretch) receptor (1)

Q: (thigh) muscle (1)

(b)



Correct position (1)

Correct labels (1)

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(c) The sensory neurone still transmits nerve impulses, so the boy would feel the tapping on the knee. (1)

Nerve impulses could not be transmitted through the motor neurone after the cut. (1)

The thigh muscle would not be stimulated to produce the response. (1)

(d) Any ***one set*** of the following: (2)

- Blinking (1); for protection of the eye (1)
- Pupil reflex (1); for regulating the amount of light entering the eye (1)
- Withdrawal reflex (1); for protection of body part / prevention of damage (1)
- Sneezing (1); for expelling unwanted particles from the nose (1)

(Or other reasonable answers)

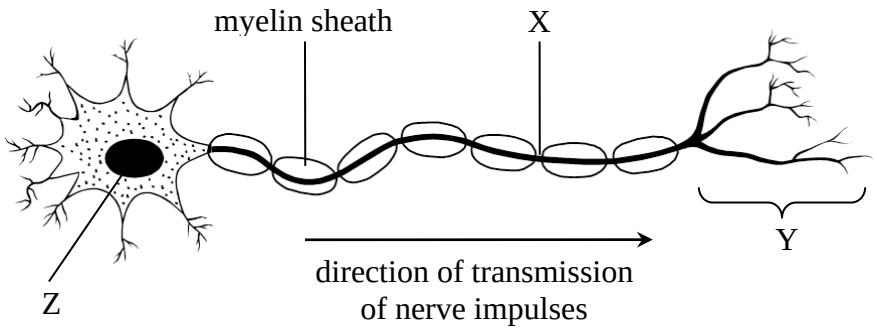
(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	2
(c)	Applying concepts	3
(d)	Understanding basic concepts	2

|!|BCELB0170000012|!| ★ **Cross topic**

The diagram below shows a motor neurone.



(a) What type of nerve fibre is X? Explain your answer. (2 marks)

(b) Nerve fibre X branches off to form Y. Suggest ***two*** structures that may be connected to Y. (2 marks)

(c) Name organelle Z and state ***one*** of its functions. (2 marks)

(d) Multiple sclerosis is a disease that affects the central nervous system. It is caused by an abnormal response of the immune system to attack the myelin sheath of neurones.

(i) State the chemical nature of the myelin sheath. (1 mark)

(ii) People with multiple sclerosis usually have problems controlling the contraction of muscles. Suggest the reason why. (3 marks)

(iii) Suggest ***one*** type of cells in the immune system that is involved in causing this autoimmune disease. (1 mark)

##

(a) Axon (1)
It carries nerve impulses away from the cell body. (1)

(b) Muscles (1)
Glands (1)

(c) Nucleus (1)
It controls the metabolic activities of the motor neurone. / It contains the genetic material for protein synthesis. (1)

(d) (i) Lipid / Fat (1)

(ii) The myelin sheath of motor neurones may be broken down. (1)
The speed up of transmission of nerve impulses is slowed down (1)
and so muscle contraction delays. (1)

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(iii) White blood cells / Lymphocyte (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Applying concepts	3
(d)(iii)	Applying concepts	1

|!|BCELB0170000013|!| ★ **Cross topic**

John went to a restaurant to have a hamburger for lunch.

(a) John communicated with the staff to order his meal. Describe the role of different functional areas of the cerebral cortex during the communication process.

(i) Sensory area (1 mark)

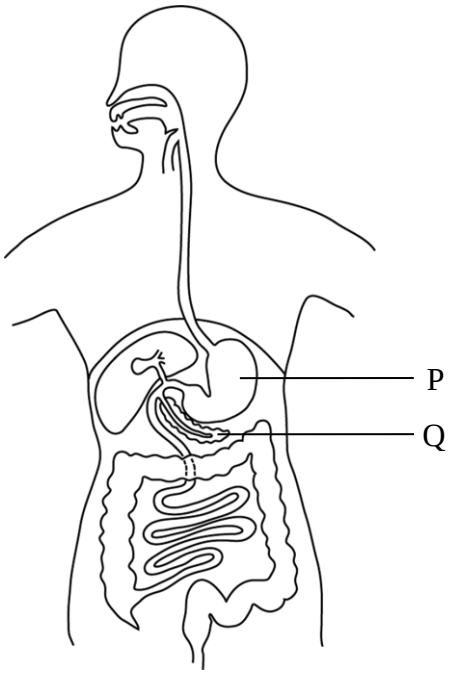
(ii) Association area (1 mark)

(iii) Motor area (1 mark)

(b) Name the part of the brain that coordinates the swallowing action. (1 mark)

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The diagram below shows the human digestion system.



(c) How does the secretion from organ P aid the digestion of the hamburger? (4 marks)

(d) Organ Q is both an exocrine gland and an endocrine gland.

(i) What secretion does organ Q produce as an exocrine gland? (1 mark)

(ii) Name ***one*** hormone produced by organ Q. Describe the action of the hormone in regulating blood glucose levels. (3 marks)

##

(a) (i) The sensory area received nerve impulses from the eyes so that he could see the menu. / The sensory area received nerve impulses from the ears so that he could hear from the staff. (1)

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(ii) The association area processed information from sensory areas and relation to past experience in order to make decisions on how to respond / what actions to take. (1)

(iii) The motor area sent nerve impulses to tongue and mouth muscles to produce speech. / The motor area sent nerve impulses to skeletal muscles to produce body gestures and facial expression. (1)

(b) Medulla oblongata (1)

(c) It contains protease (1)
which aids the digestion of proteins in the meat into polypeptides. (1)
It contains hydrochloric acid / It is acidic (1)
and this provides an optimum pH for the protease to act on the proteins. (1)

(d) (i) pancreatic juice (1)

(ii) Any ***one set*** of the following: (3)

- Insulin (1)
  It stimulates liver cells to convert excess glucose to glycogen for storage / stimulates body cells to take up more glucose from the blood (1)
  to decrease the blood glucose level when it is above normal. (1)
- Glucagon (1)
  It stimulates liver cells to convert stored glycogen to glucose (1)
  to increase the blood glucose level when it is below normal. (1)

(12 marks)

##

	Skills assessed	Marks
(a)(i)	Applying concepts	1
(a)(ii)	Applying concepts	1
(a)(iii)	Applying concepts	1
(b)	Understanding basic concepts	1
(c)	Applying concepts	4
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	3

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|!|BCELB0170000014|!| ★ **Cross topic**

Tony went hiking and saw a snake coming towards him. He turned and ran away quickly.

(a) Describe the change in the lens of his eyes as Tony saw the snake coming towards him. Explain how this change was brought about. (3 marks)

(b) Describe how the nervous pathway that brings about Tony's response of running away. (4 marks)

(c) Is the response of running away a reflex action or a voluntary action? Explain your answer. (2 marks)

##

(a) The lens became thicker / more convex. (1)
This is caused by the contraction of the circular ciliary muscles (1)
which leads to a decrease in tension in the suspensory ligaments. (1)
The lens becomes thicker due to its elasticity.

(b) Photoreceptors in the retina of the eyes were stimulated and they generated nerve impulses. (1)
The nervous impulses travelled along the optic nerve to the visual centre of the cerebrum. (1)
The cerebrum made decision and sent nerve impulses. (1)
The nerve impulses travelled along the spinal cord and motor neurones to the muscles in his legs, causing the muscles to contract. (1)

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(c) Voluntary action (1)

because the cerebrum is involved. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	4
(c)	Understanding basic concepts	2

|!|BCELC0170000015|!|

(a) What is the difference between an endocrine gland and an exocrine gland? (2 marks)

(b) The pancreas serves as both an endocrine gland and an exocrine gland.

(i) Describe the endocrine function of the pancreas. (2 marks)

(ii) Describe the exocrine function of the pancreas. (2 marks)

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(c) Apart from the pancreas, give another example of endocrine gland and its product.
(2 marks)

_____

_____

##

(a) An endocrine gland secretes its products directly into the blood. (1)
An exocrine gland secretes its products into ducts that lead to the target tissue. (1)

(b) (i) The pancreas secretes hormones (insulin and glucagon) into the blood (1)
to regulate the blood glucose level. (1)

(ii) The pancreas secretes enzymes through the pancreatic duct into the duodenum (small intestine) (1)
to help in digestion. (1)

(c) Any ***one set*** of the following: (2)

- pituitary gland (1); growth hormone / anti-diuretic hormone / follicle stimulating hormone / luteinsing hormone (1)
- thyroid gland (1); thyroxine (1)
- adrenal gland (1); adrenaline (1)
- ovaries (1); oestrogen / progesterone (1)
- testes (1); testosterone (1)

_____
(8 marks)

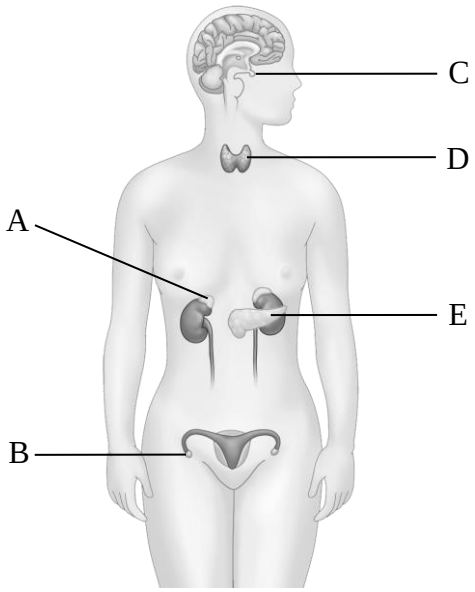
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

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|!|BCELC0170000016|!| **Cross topic**

The diagram below shows the main endocrine glands in the female body.



(a) Complete the table below. (4 marks)

Gland	Hormone produced
B	
	glucagon
D	
	follicle stimulating hormone

(b) Describe the role of glucagon in regulating blood glucose levels. (2 marks)

(c) Apart from the endocrine system, the nervous system also coordinates body activities. Describe how the endocrine (hormonal) coordination and the nervous coordination differ from one another in term of the nature of messages and transmission routes. (4 marks)

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##

(a)

Gland	Hormone produced
B	oestrogen / progesterone
E	glucagon
D	thyroxine
C	follicle stimulating hormone

(4)

(b) It stimulates the liver to convert stored glycogen to glucose (1)
to increase the blood glucose level when it is below normal. (1)

(c) The endocrine coordination makes use of hormones which are chemical in nature. (1)

The nervous coordination makes use of nerve impulses which are electrical messages and neurotransmitters which are chemical in nature. (1)
Hormones are transported by blood. (1)
Nerve impulses are transmitted by nerve fibres. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	4

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